

A convergence of TAME (Levin 2022) and Active Inference FEP perspectives on dissociative experiences

Source: [YouTube](#) Playlist: 4th Applied Active Inference Symposium (2024)

Duration: 57:53 | Uploaded: Unknown Transcript method: auto_caption

all right welcome back here with Alexi tolinski a friend and colleague and collaborator on this paper temporal depth in a coherent self and in depersonalization theoretical model so go for [Music] it well uh I'd like to say hello to everyone and thank you for being with us after such a long and fruitful day at uh 6: p.m. on on the east coast and uh it say I appreciate your interest and thank you Daniel thank you active inference Institute for this fantastic Symposium um so uh uh we'll be working talking today about this paper on dissociative States and it's a collective effort uh this paper is in peer review right now uh early uh preprint was posted uh on the site and it was uh a work by Michael Levan Chris Fields lanot D Costa Rachel Murphy Daniel who is here uh David Pinkus and myself and um this is the link to the paper if you'd like to take a look at the earlier print uh perhaps some of you would like maybe to get a feel for what we're talking about and this is not the patients account this is a a poem uh from uh Kurt V and his song Pretty pumpkin I woke up this morning didn't recognize the man in the mirror then I laughed and I said oh s me that's just me then I proceeded to brush some strangers teeth but they were my teeth and I was weightless just quivering like some Leaf come in the window of a restroom um and this uh poem was cited by Paul moleno he's a specialist on did in England on the Psychiatry and Psychotherapy podcast you can also find patients account on YouTube I couldn't post the video here for copyright reasons um so dissociations are a spectrum and they range from mild to severe transient uh to Chronic they can be benign and pathological and certainly they're not a bright line category uh the causes vary for them as well we call it iology in the clinical World they can be caused by psychological trauma neurological conditions substances seizures anesthesia and uh different sleep phases there's a subset of conditions here we call dissociative disorders typically in the history of these patients we see prolonged elevation of stress accompanied by repeated traumatic experiences in circumstances where they feel There Is No Escape typically

childhood abuse and neglect and we call that complex trauma which has a different presentation than acute trauma and we'll touch upon why this specific pathway of etiology is important for dissociative disorders um well I'd like to say why this matters what's at stake what's the motivation for this paper and we have many authors and they have variable motivations but I'm a clinician I see patients all day and um uh for me the main reason for that is these conditions are hard to diagnose many patients go undiagnosed for a long time they're very hard to treat uh and we're sometimes content with partial results and the risk of self harm or suicide is higher for these patients and there are many many reasons for that uh we think that perhaps one of the reasons is that there is no unified theory of dissociations we have kind of um neurobiological ideas the corticol limbic inhibition hypothesis we have uh psychodynamic ideas we have uh attempts to model dissociations from first principles by Anna and other colleagues but uh these different attempts are not tied together and certainly not tied together with the patients uh subjective experiences so we are trying to make steps in that direction and we don't claim that we've built a unified theory of dissociations but what we're doing here is we're focusing on one aspect of dissociations one focal point that is a collapse of the temporal depth in any kind of dissociation regardless of the cause by temporal depth we mean how far into the future the person can plan what we're hoping for is that this uh uh focused approach has the potential to uh allow us to to to um perhaps develop uh more accurate diagnostic uh uh procedures and and to possibly um help increase efficacy in in therapy with full respect to how difficult complex and messy this work is um uh additional point of convergence is we explore multiple theoretical perspectives in the paper and they all converge on the same um points and here is the main thesis um the collapse of the temporal depth is necessary and sufficient for the onset of a dissociative episode so we're going a little bit further than colleagues who already mentioned a relationship between temporal depth and dissociations Dean and Anna Yunik and others and Carl Friston were saying it's causal that there is no dissociation without the temporal depth collapse and the collapse of the temporal depth will reliably lead to dissociative episode both we are um uh saying that temporal depth collapse can be conceptualized as a common pathway for the onset of dissociative symptoms regardless of etiology and finally we suggest that the focus on restoring the temporal depth can be an effective strategy in therapeutic interventions one of the theoretical perspectives we explored in the paper is Michael Iltis's framework technological approach to mind everywhere which I think is very useful uh and unfortunately I haven't seen many uh attempts to uh use it in clinical applications this is the title and the link to the paper uh there's so much

in Michael Len's work that is useful uh his uh paper on cancer which is based on his experiments is also one of The Inspirations for our paper here but here's another inspiration if you dive into Mike's work you'll probably see uh what he claims based on experiments that all intelligences are collective intelligences however ever at our subjective level at the level of the conscious mind we don't feel that way uh we perceive unity all our percepts Sensations thoughts feelings come together in a cohesive and coherent whole a single conscious experience in the here and now and uh we will talk about this seeming uh Paradox uh if you wish in in our paper um some of the core tenants of the team framework not all of them but some key points uh commitment to graduality the there is no bright lines uh Michael suggests that intelligence is built gradually and uh by accumulation of complexity material Independence uh our model is not based on neurons or central nervous system we will talk certainly about a subset you know I'm a clinician I work with human beings so we'll talk about brain-based models but generally the model applies to synthetic agents and cells that are not neurons commitment to empirical uh research as opposed to just exchange of opinions on the subject um Dynamics the C are not fixed they're malleable and plastic and cell can be nested and overlapping cooperating and competing both laterally and across levels uh this is an illustration of one of the key Concepts in Len's framework um uh called a light cone and what you see here depicted is various organisms in the middle of the slide you see a tick and a dog a human being possible AI by compound intelligence such as an end colony and you see this sort of nested and embedded structure on the picture on the right and L cone has two uh Dimensions there's a spatial Dimension and and a on the horizontal plane and the vertical Dimension is temporal and TI for example is not particularly concerned uh with things happening a mile away uh it cannot recall far from the past and plan far into the future well human beings are capable of uh planning things until uh uh that happen after we die a living will and uh with the use of language and culture and books we can uh use things that happened before we were born so we're capable of huge light cones and I'm highlighting this word capable of it doesn't mean that in our wakeful hours alert hours we are walking around with the significant light cones they contract and expand the depending on various circumstances but we're certainly capable of them while ticks or not and the vertical dimension of a lyone is exactly what we mentioned before that is the temporal depth there is of course a perspective on the same thing in fap and I like this very Su phras by Carl friston who said because consequences follow causes the generative model acquires a temporal depth and more formally uh some agents are only capable of modeling their immediate environment and in the

Here and Now While others are capable of planning actions into the future and the generative models capable of planning are referred to as temporally deep generative models so temporal depth under fep is the length of the temporal Horizon that is considered during planning uh we're moving on to describing the if you wish healthy self um and we do mention that it is a materially independent um structure where you know in principle we can talk about of the selves of synthetic agents and you know cell collectives uh uh selves depend on an organism's ability to infer and here we will uh provide uh a viewpoint on the two clinical terms familiar to clinicians such as derealization and depersonalization so the organism's beliefs about the world are the inferences about the external environment and the Continuum of issues in this process we will call derealization while the organism's beliefs about its own internal milieu its body and mind we will call the self and the Continuum of issues of this process we will call depersonalization the S are hierarchical composite nested and with boundaries uh so we consider self to be a system consisting of hierarchically structured beliefs which effectively makes it a dynamic hierarchical generative model and taken in its entirety the self is informationally separated from the external environment by a mark of blanket or near Mark of blanket by boundary would like to emphasize I apologize if this is Trivial to the audience of the active inference Institute that this boundary is not Material like skin it is informational uh through which the self and not self interact so to emphasize the self as we have defined it is a model it is not an organism the organism constructs and implements it so the self is quite literally the construction of the mind this is an illustration of that from early work by Michael Lan I think this was his doctoral dissertation materials and what you see here is um Duc embos that started from oite and then you have blastoderm you after cell division you have a collection of cells and at the top left if nothing is done then this blastoderm will develop into a single ducky you see this green shrimp looking thing that's a single duck however if you introduce a scratch in this blastoderm and the wound heals the same exact cell Collective May develop into twins or triplets you see the red yellow and green duckies now think about that for a moment what happened here um this Collective of cells computed what is me and what is not me and the hardware did not change but in one case it made a decision that this is all me and the rest of it is not me and another one it says there's three mes so you can see that this is not Hardware um s are also as I mentioned hierarchical composite nested and they have boundaries so now we're moving on to components of the self uh that are each component is separated by their own boundary and Mark of blanket and because we're using the word hierarchical uh we will uh Define that there is a core self a core component if you'd like a

visual metaphor that Mark SS uses we can think about an onion and there's different levels of the layers of the onion and there's a core uh in the middle we will talk about that as applied to humans and mammals and rodents so now we are in organic uh in organisms uh this term is from effective Neuroscience from Yak pip and Lucid Bin's book um and also you know I believe SS and Terrence used it uh so core self is non-reflexive non-reflexive and dominated by raw feelings uh and part of purely effective core form of Consciousness so the effective homeostatic mechanisms allow an animal to problem solve in novel environments and I'll give you an illustration from Mark SS um to maybe just um bring it home uh imagine that I'm in the room full of carbon monoxide and I've never been in that before uh so I don't really know what to do but I feel badly I feel badly in a very specific way there's air hunger Suffocation distress uh and let's say I'm moving around the room but if I move to the left and I feel a little bit better and that's information and if I move to the right and I feel worse that's also information perhaps there's a door on the left and there's a draft underneath the door and I feel better so this Compass this internal mechanism the homeostatic mechanism allowed me to find out a possible solution and um we have a whole collection of these and to mention you know the actual hunger uh for food and thirst and social homeostats such as rage and fear and panic separation distress Etc so the collection of that is what Panksepp and SS conceptualized as the core self uh uh therefore the set of predictions in the core self that all the life sustaining effects will be at or near their settling points which means in emotional terms contentment all my needs are met I am okay thank you very much right and that is the minimum of the variational free energy if there are deviations from it such as hunger or thirst or uh air hunger then that would be a prediction error accompanied by a negative feeling like in the example that I used above uh in mammals we're not putting the equal sign but we're describing the anatomical structures that are uh participating in uh serving this function we're talking about the upper brain stem regions including the reticular activating system and the periaqueductal gray they mediate the function of the core self and you can see a lot more details in Mark's book Hidden Spring a journey to the source of Consciousness so um or self is thought to be present in animals without the cortex such as decorticated cats that are actually more effective and not less and the cases of human children born without a cortex hydrocephalic children they have emotions when you know uh you put a brother on on the girl's you know uh chest she smiles and she cries so the core self does not require a functional neocortex the higher levels of the brain structures in humans including the cortex mediate the higher levels of both Consciousness and the self and sometimes it's referred to as extended Consciousness which allows us to reflect on what's

going on such as to say I feel anxious right now uh there is an a symmetry in this uh um model where the higher levels the peripheral levels of of the self and and Consciousness cannot exist without the core um while their verse is not true the core can exist without the higher levels one of the other illustrations uh of that which PM cites this is based on Fisher's work in 2016 is that a tiny lesion a 2 cubic millimeter lesion in a parabal nucleus will reliably and IR irreversibly reduce a coma the patient will never become conscious ever again while you will never have anything like that happen in the cortex you know this small lesion in the cortex will not result in catastrophic outcomes like that and this is one of the reasons why neurosurgeons don't usually like to operate around the brain stem this is too consequential too risky so everything was said about Consciousness the core and extended is applied to self as well that peripheral self components cannot function without the operational core self and we will illustrate that with specific examples in humans later on and addition to this onoff where the thing is completely operational or not we can talk about the influence and the core self has the ability to change the regime of functioning in other more peripheral self components by inducing phase transitions in them and the Brain mechanism for that is generalized arousal um when the higher levels of consciousness are present and functional then we have representations such as fruit and vegetables and you know in the core self we have Raw effect such as I don't know separation distress or hunger but with the higher levels of consciousness presence we are capable of uh uh thinking things like I want an apple or the aect the drive is in Freudian language and in SS language is attached to the object and similarly we can have these constructions about the meta object such as I uh such as I look pale or I am a pessimist and I here is a mental object and therefore I is a metacognitive construct an abstraction and it only exists at the higher levels of the self such as autobiographical self it is not used or even needed in the core self to reiterate core self as a system has the capacity to detect uh the effective prediction errors and uh attempt to minimize the variational free energy through action without any eye we're moving on to peripheral self components and we'll describe only a few uh and the terminology here is different from what you may read in wonderful book by Neil Seth being you so the first one is autobiographical self and that's a system dynamically representing our life's history if you for example remember your 10th birthday and the cake is on the table and Mom is there and your friends are there and the sun is shining and everyone's smiling and it's Tuesday that's an episode that's an episodic memory and a collection of them organized through time is our sort of life narrative um there is a bordly self which is a system dynamically building inferences about our body and various representations and interceptive

processing there's a social self which is a system representing inferences about how we are seen by others and how we present ourselves and act in the social environment each peripheral self component such as bodily self for example has its own Mark of blanket and each component is embedded into the whole self and also contains subcomponents creating a nested architecture that you saw in the T light cone picture before about the unity that in humans at the higher levels of the self such as at the autobiographical self level uh uh uh these these levels create an impression of a unified experience and this is just the experience of the autobiographical self not the entire system the autobiographical self claims to itself and others to be the entire self while it is not so Anil Seth refers to these things as delusion we have a delusion of being a unified I don't know alexity right but you know what we're saying is that it is that system that is deluded um uh the autobiographical self subsystem I'm ambivalent about using the term diluted I think it it's not the optimal one because we use it in the clinical world and I prefer it uh called inference or belief that we're unified State differently we can say that a stable belief that I'm whole in the um I'm complete in the autobiographical self um its generative model contributes to us feeling as a mtic self therefore the subjectively perceived Unity the coherence of the self is an inference related to coherence another thing we can talk about is continuity and nearly is a theology stemming from the definition of Mark of blanket our self remain the same persistent time while all the processes or Communications across its Mark of blanket remain functional and we can um offer you another viewpoint on the same exact phenomenon from if you wish more neurobiological perspective or psychological perspective if you recall maybe you're at the train station or you know anywhere in the city and you see someone and you think you recognize them but you're not sure you know this sensation of recognition of familiar uh so the familiarity novelty calculations is what we're talking about here and depersonalization is a breakdown in that something that is familiar myself suddenly abruptly becomes very novel and not recognized but actually these calculations neurobiologically are not uh bright line dichotomous categories they are probabilistic and they are Spectrum but we course grain them in the mind often into yes or no familiar or novel so that subjective impression of yes I know this person or not is just the course grading of the continuous variable representational capacity is important uh the Deep generative model implies that the agent is capable of planning into the future which in turn requires an ability to generate store and retrieve counterfactual data generate store and retrieve as memory so therefore when the representational capacity is completely or partially impaired for any reason the agent completely or partially loses an ability to infer leading to

depersonalization and der realization if we talk about health and pathology briefly the two key terms here are voluntary and involuntary you know again all of us during our day we can either sit down to plan things you know for you know 10 year temporal Horizon which expands our temporal uh depth or when we uh meditate or uh rock climb you know then we can be more in the moment in the here and now but uh trauma doesn't ask per I mean perpetrators are you know there's nothing voluntary about it and so that's an involuntary collapse of the temporal depth uh which usually is leading to pathological conditions this is a perspective from the nonlinear dynamical systems you know if again I am in uh you know adult healthy alert functioning and I may have small deviations from I IM me say I'm daydreaming or something like that and the collection of these points from which I rely ably return to ime can be described as the attractor Basin and the whole thing in terms of attractor landscape is the set point attractor regime ime and ime would be the variational free energy minimum and that is a fairly stable system perhaps in the autobiographical sum right now in order to for that system to change some energy uh needs to be applied and with external interference of sufficient power this attractor landscape can change when that power is applied temporarily there's a phase transition the attractor landscape evolves into something else perhaps during that transition you have a chotic regime and if we are in the case of trauma then the functioning can be chotic for a while but then it may settle down so to speak into one of the possible post-traumatic presentations there may be multiple postraumatic presentations and one of the possible options is depicted here this is not again Universal but you know it's one of the possible options what you see here is a multiple coexisting uh Point attractors represented by local Minima of the variation free energy and they can correspond to Alters in D ID uh and in this example you also see that they're surrounded by a global minimum variation of free energy now also if you look at the uh area the region between the local Minima then in dynamic IAL systems that's called a repeller and uh repellers are uh notoriously chaotic so uh that suggest that you may not have very good predictability of how you will transition from alra X to alra y or alra z uh to say a few more words about this when we switch between the altars then we are in regime of itinerancy this is the term Carl Frist likes to use a lot and you know justifiably so also when we have transitions from X Ultra X to Y that the time is disrupted you know in the idea you don't have this flowing continuity of time there's a series of disruptions effectively then your uh coherent large light cone is broken into uh pieces here now imagine that this patient with with this condition reached out for help and they want to get better but this uh attractor landscape already acquired some stability and

so again we need some interference to destabilize the regime the difference here would be that in therapy in the process of healing the this uh destabilization of the pathological attractor landscape is voluntary controlled and guided ideally by a licensed and competent mental health professional and uh that's that's the uh the difference from the onset of uh acute trauma so Psychotherapy can be once such intervention and it's the treatment of choice we have very limited efficacy with medications and now it's very uh uh uh kind of um fashionable to talk about augmentation with psychedelic treatment or perhaps neuros stimulation it's important to say that just destabilizing that landscape doesn't guarantee anything at all so when people present this idea that you just go to like I don't know and have some iasa or anything ketamine infusion and you're better is a myth because you know the important part is that the patient settles back into health and just the destabilization of this landscape doesn't lead to that you still need uh therapy you still need you know to process the trauma not just to destabilize the pathological regime um and that therapeutic process when you destabilize the the pathological attractive landscape is another phase transition temporarily temporarily chaotic with the idea that eventually it leads you to a healthier functioning either a attractive landscape corresponding to cohesive uh system or at least a little more coherent system with less Alters partial Improvement let's talk a little bit more about the phase transition so before things happen in in in health we have a nested and embedded uh self and that can be described in in FB is a hierarchy of coupled interconnected attractors car friston you know uh wrote that slower attractor can possibly control the phase transition of the faster attractor and in our system the core self is is feeling based it's effective and it's one of the components uh of uh effects of feelings is generalized arousal and we know that the CH changes in generalized arousal level can lead to the phase transitions of the entire neuroc cortex from periodic to chaotic State and back so this is one of the possible uh kind of you know uh explanation in our model how this works the phase transitions um you know a few more uh um thoughts about it trauma is not a um guided transition it's an effective storm it's a flood of cortisol that is abrupt and severe and that uh abrupt increase in generalized arousal leading to the destabilization of a healthy attractor landscape in the autobiographical self but then when there's an onset of a dissociative syndrome after trauma then uh the regime settles down to a lower energy uh perhaps effective numbness that patients talk about and in a Psychotherapy not immediately but when the patient is ready which can take a year or more you know when there's active trauma work then there is again an increase in in energy level with with the idea eventually to settle down to a healthier regime of functioning a few clinical observations starting from

dissociative disorders which are conceptualized traditionally as uh defensive response to prolonged and inescapable trauma clinicians talk about uh and patients report this this quality of emotional flatness or deadness or numbing sometimes they call it freeze so in our model it corresponds to issues with Communications between the core self and the peripheral self components for example that autobiographical self in this regime operates as if it is uninformed by the vital emotional informational flows that reaching in the core self one of the possible ways it can be achieved not the only way but one of them is the frontal prefrontal cortex inhibiting the lyic structures that's the cortical lyic inhibition hypothesis and in fep terms that means lowering of precision associated with effective prediction error messages um we're moving on to also severe dissociations but temporary such as general anesthesia propol and uh uh uh it's a very found change in the body that influences the Gap Junctions between the cells and recovery from anesthesia is a very interesting process Michael Lan is fascinated with it it's actually a little bit of a miracle that we come back to be the same person we have been before and some people don't uh you may wake up a pirate uh and uh these conditions are called post-operative delirium dissociative Amnesia post-operative cognitive dysfunction this is also an illustration itself is computed and plastic and malleable it's not hard um you know Hardware um and perhaps we get oriented with the use of our memory systems including the episodic memories another severe case is a seizure where not only autobiographical self is impaired but also the social and the bodily self and one of the reasons for that is the necessary resources for the functioning of the self components are not operational in seizure uh moderate dissociations uh in neurological cases but permanent ones are for example the perhaps one of the most famous neurological patients hm who had a bilateral hpoc compus damage uh and uh in Britain there's Clyde wearing and Anil Seth describes Clyde wearing in his book being you this patients do not have episodic memory encoding and effectively they live in about 10 minutes of the here and now uh there is no past and no future beyond that and if you read Clyde wearing's diaries that he talks about the ser of Awakenings he said oh I just woke up and 10 minutes later oh I woke up and then I woke up and if you show these Diaries to a clinician who is an expert in did they will which I've done you know I talked to Roxy wolf one of the fellows of the international Society for the study of Trum associations and she said that you know this patient looks perpetually depersonalized there is no continuity itself Corso patients they don't recall the past they can fabulate the past and in Alzheimer's in early phases there's issue with recent autobiographical memories but some access to the remote autobiographical memories all of these cases are discontinuity in

autobiographical self um uh more frequent perhaps cases or substances ketamin I doses of THC or iasa lead to depersonalization and dualization and importantly if you read the studies then these patients actually report they tell you that their sense of time is impaired not only that the planning is impaired as well um if you have a alcohol blackout and the person tells you on Wednesday that they have no idea whatsoever what happened on Tuesday after 6 then uh this is a discontinuity this is a gap in episodic memory uh there's also something called traumatic Amnesia there very said cases of high school shootings where you know the person literally doesn't remember you know 15 minutes of what was happening in the cafeteria what happens in acute PTSD is that you know uh hippocampus is reaching cortisol receptors and there's a flood of cortisol and the effectively the episodic memory of trauma the cohesive context contextual time stamp memory is not laid down in a stable fashion you have shards of glass you have phobic memories semantic memories procedural memories but not the episodic one which is also discontinuity autobiographical self and that is involuntary uh and we talked about mild benign everyday uh dissociations uh next time you wake up you know maybe tomorrow the day after you know try to monitor how what's going on uh and you may catch that pH of what met singer calls you know a non-ic self where things are just flowing through your mind but your eye is not fully out borous yet uh to summarize the level of consciousness regime of Consciousness pathological States and all the underlying resources necessary for the successful operation of each self component collectively influence the complex dynamics of the system um a neurobiological perspective briefly the historically most famous one is uh cortical limic inhibition hypothesis by Sierra and barus who postulated that uh prefrontal cortex is uh inhibiting the lyic structures including the amygdala and the interior singular cortex ACC the subsequent studies don't fully support that hypothesis but partially support it I like the work by Ruth lenius and her colleagues and guess what you know her uh revision her refinement of the cortical liic inhibition adds Dynamics transience context and itinerancy all the things we like uh uh uh they talk about two regimes even in patients with dissociative syndromes over modulation and under modulation ation again itinerancy in not one regime but multiple regimes over modulation is indeed hypoactive amigdala and hyperactive prefrontal cortex which is described above but also these patients less frequently but but you can see it in a reverse regime where amygdala is active and and PFC is less active uh uh over modulation dominates in dissociative uh conditions they refined the anatomical Network and talked about different sub regions of amygdala the central medial amygdala basolateral amygdala dorsolateral periductal gray vental lateral periductal gray Thalamus and other regions and this network of

regions collectively. Works to create specific post traumatic regimes they also described uh the regime where patient has access to painful memories usually uh is accompanied by under modulation very active state with amigdala active and uh when they don't have access to painful memories they're more in a dissociative State we can also briefly talk about what are the anatomical substrates and the neuro psychological substrates of the temporal depth collapse and again acute trauma hpoc compal damage and dementia are affect the episodic memory system uh if there's an insulin damage then you can expect the disruption in the bodily memories and perhaps in camine use you may have a dysfunction of prefrontal Cortex and frontoparietal Network which renders the work in memory uh of the patient dysfunctional you can imagine other circumstances when you have PFC in front of paral network impaired that usually leads to a psychotic state where you know it's very hard to think you know we need work in memory to be able to think um about dissociative disorders again the point made at the very beginning that prolonged inescapable exposure to highly stressful environments can be seen as lasting stress beyond the agent's ability to manage which results in the breakdown of um tame light into the uh components a fragmentation decoherence and discontinuity when the landscape uh attractor and repell Landscape corresponding to a pathological condition takes root and in contrast to that if you look at the single uh episode of camine intoxication then you have transient working memory disruption leading to transient dissociation again to summarize lasting or temporary severe or mild dissociative experiences are accompanied by the collapse of the self temporal depth and we are ready for discussion which I very much look forward to thank you Lexi great presentation thank you okay where should we begin what would be interesting or or given where the work has gotten over the last few weeks where what do you feel like is most relevant well perhaps any colleagues have comments I don't know if Jus is with us and he's a esteemed neuros psychiatrist and he sees a lot of patients and perhaps neurological patients and he can comment on some of the things we've mentioned I I had a uh clinical meeting with with uh colleagues who do a lot of dissociative works and I mean I'm seeking feedback and critique and so whatever ever people have whatever thoughts people have yeah if there's anyone in live chat they can write a question a few pieces that stuck out differently in how you presented it uh in comparison from working on the paper was the narrative self's role in the maintenance of The Narrative of unity and about what is the experiencer of that or is that all it can do why does it do that why does the autobiographical self have that kind of a voice or a cognitive function is audio working okay can yeah can we hear each other I I think why is a tricky question you know it might also

be tell logical question I would say that it's probably functional you know uh for us to have coherence elves uh and um I don't know why uh but um we are observing it so we have descriptive level we have observations that I feel like the same Elixir and it's a little bit of a miracle that as we know that the cells in the organisms they are they die and they're born but I still feel like the the same person even so uh we change developmentally I'm not the same person as I have been at age 5 or age 15 but I still maintain this story if you wish this believe that I am the same and um I also protect the internal milieu from external threats you know of the same organism and if we look at other species say an oak tree that you know uh grows from an acorn it is still the same Oak it's the same organ uh so um I I perhaps in biology as well there's a need uh for the system inside the mark of blanket to maintain that saying whatever is inside the mark of blanket is the same me but it is a belief you know and uh um I don't know if I'm answering your question maybe I'm just free associating to it so yeah is that a clinically valid method um well it it reminds me of mortal computing and there's a lot of focus on how behavioral activities in a niche relate to the material stability and I think you really drew carefully on Mike Levan at all's work to talk about the analogous Dynamics on a landscape of a narrative self and that's just kind of that was like a trampoline but then you also complicated that with multiple elves like the social and the narrative and so then there are these multiple disassociative or like it's kind of they each have their own unity and yet they can be discordant and unified and so then is there any general framework for different possibilities open-endedly of different regimes these are great questions or even maybe challenges to develop it further but I agree with you if we look in the bodily self right I don't perceive my hand to be a separate entity I believe I I feel for it to be part of the whole organism myself my body right and so in the bodily self there's also a level of unity I'm just saying that this is again a computation within that specific sub system that my body is one whole self and we can see how that gets uh perhaps distorted in somatoparaphrenia or even rubber hand illusion where uh this hand is perceived as not you know so uh and so we see that the system is plastic and malleable and and prone to change but and when when no uh interference is happening it continues to be cohesive is one body and one mind you know this autobiographical I I've noticed you Ed the term narrative uh self I think that Anil Seth's term and I prefer autobiographical for the reason that uh rats they have episodic memories and they don't have our verbal abilities um uh and so they they also have episodic memories which are what we when the contextual memories right and I think that uh there's the same eye in all autobiographical memories and that's unity uh yeah yeah thanks you a few it's it's it it it

probably is more accurate to say it's a narrative cognitive capacity that encompasses autobiographical or Alo biographical type narratives um and I'm reminded of uh NPR radio discussion from a few days ago that I listened about Memoir and about what Memoir was and could it only be written when the person did or didn't change a lot but what or who doesn't change in every single one of these ways so then what would that mean like what so that that that was kind of a funny connection and then and then um do you have any thoughts on that or have another active comment no I'm just enjoying your comment I don't know if I have a a I thought on that yeah you you've see many clinical Memoirs and presentations of self-understandings and all of this so there's just so many and it gets brought out so sparsely like and indirectly and it makes a lot of sense to the person who might be saying it but that that could you could be too or not confident enough about that interesting you brought it up thank you for that Daniel I think that you know when we theorize say in Dynamic theories then we uh usually think about different sub agencies of the mind like Freud for example in his know structural model he had the Eid ego and super ego where I want pistachio Asam is the Eid and pistachio es is fat and calories that's the super ego response and stash esum is \$4 and you have three and a half is the ego uh but there's a whole bunch of them we can name a sort of a sub agency of an internal judge you know which is sort of super English or a neutral Observer or uh an accountant who's running Itali or and all of this so we all have sort of a jury there of different kind of regimes of operation if you wish but uh the patient is often surprised because when they for example self- criticize and you talk about that in therapy and say do you notice the voice of your judge it's kind of you know uh not very fair I mean there's no misdemeanor everything's a felony and judge is present all the time like no it's not it's me I'm doing it you know I screwed up you know so again you know they default regime is is me myself and I that I am you know I am bad and when we we discuss these things and we open it up and we show the different kind of uh we show a theater with different voices or a jury with different representations it's usually kind of an Insight it's a surprise but that allows a bit more control um for the patient in clinical work soon okay I'll read a question in the live chat Pablo FM wrote thank you very much I find it super interesting I wonder why humans seek for drug experiences having been so decisive in human history coffee could be a great example tea tobacco alcohol have also been very popular among civilizations it's a great question I don't know if I have a good answer it's a huge question and there's a good paper you can find it on Google's caller a pbet by me zelner Mark SS and other colleagues I think Yak pip was alive it's called why depression feels bad and what do addicts really want and it's based on effective

neuroscience and what it postulates is that uh of course there are individuals who have multiple drugs that they are you know using all the time so there's an upper and the downer and other things but there are usually some innate kind of predispositions where people who are wound up in tense they sometimes crave alcohol because it's uh binding to the Gaba receptors and it's sort of you know sedating us a little bit it makes you a little more less kind of wired up people who are very painfully lonely and uh uh detached and they crave human uh comfort and you know to feel together and loved and accepted they may if they discover opioids that would be a you know lock and key match that would be a dangerous combination um and uh perhaps when somebody's oriented to Performance and they're very bored all the time and they want to something you know to do better and more they may gravitate toward uh stimulants and so there's these things uh and there's some common factors among different agents uh such as you know I believe that multiple substances they have the dopaminergic effect and you know a lot of motivations are subjective in individuals I've had patients say that I'm smoking pot to slow down time I've had patients tell me that you know p is my oldest friend I've been doing it long longer than any person that I've been with uh some people seek this state of numbness and they want to kind of not feel as acutely they want to sort of feel desensitized so these states of dissociation can be desired um and that's briefly I guess what I what I wanted to say about it so well thank you I'm reminded of a comment from ma in the live chat earlier this morning and she was kind of pointing to how one mode of how we've seen active inference applied in these presentations is looking like athology at the sequences of behavior but not really looking at the self-maintenance as a behavior like especially talking about digital agents or or that it's just about what do they perceive what do they do it's taken as a granted or a separate question how they exist yet also there's this self-maintenance self-reconstruction character that doesn't even need to have anything to do with the athology its Dynamics can just be like morphologically playing out not necessarily doing some cognitive task out in the world in any particularly effective way yeah it's a very interesting thought and for some reason I'll allow myself the Liberty to free associate again it reminded me of yal Harari who went interviewed he said you guys go to the gym You observe your diet and you take care of your body and he said I take care of my mind I meditate an hour a day and uh it goes to silent Retreat a month and a year and the interviewer challenged him I think he was from The Daily Show he said how do you manage that and her said I got no kids you know so but in addition to that he's also in therapy so there's a uh that idea of really maintaining the mind and not just the body with behaviors we're all oriented to what do I do do do what do I

think think think which are sort of verbal virtual behaviors but how do you feel as my great colleague Raji uh Warrior says we're human beings not human doings we have a very tough time being we're all about doing more that's true and so uh yeah and also Harari talks about informational hygiene where a lot of stuff in the social media is noise and trash and a lot of it is consumed and we have pollution but you know so he doesn't do you know he reads books and he thinks in Millennia but I don't know I mean I went sideways from your comment but it's just that's what where my thought went so yeah it it seemed like a very relevance when people are discussing the the the unity or the articulation of Mind and Body which is kind of what this unified promise on the cover of the textbook suggests is what what would it be to have a unified model of brain mind behavior and so on then the the analogy or the connection with the physical and the cognitive is just it has to be taken as a starting point not like a hot take yeah yeah but what you just said reminded me when I read Michael Len's work and got more familiar with it I actually had regret I'm like why were we not taught that that was a bit of a revolution in my mind it was a big pivot because if you go through graduate program in psychology it's all about the unity you know I am the same you know cohesive and coherent and the fact that we have intelligence at the level of individual cells and then the cell Collective and then tissue and organ and all of this thing that I think is one montic Alexi is a pyramid with Communications bottom up and top down and laterally is is a revolution I think that you know I I wonder what models of psychology can be built with that and they don't exist yet it's all about one big block and what happens to it I am depressed you know there's no collectives this is one big block right yeah it's a really important Insight I think that's the kind of frictions and and experiences will start to learn more about and see produce in different ways like just seeing how you connected the topics to specific quotes translated the ontology re worded descriptions using kind of like a translation sometimes really just as simple as a lookup of words just authors using one word and being able to see an otim onology term equivalent or roughly related it's uh and and what what does that look like or I guess as your last comments what do you think after to the paper after this now like what are the directions towards which people would say okay that's applying active inference in a clinical setting I my goal eventually I think is to get to Therapeutics and for example you know um actually I probably will not use that example it's it's a good material for the second paper that we'll follow up with uh but I I do think that you know the question is what do we do in therapy differently how do we understand how do we formulate cases differently with this information I I haven't heard it clearly said out loud in the did literature there's a lot of perspectives there's

a lot of endless complexity and there's a lot of nuance but if we can highlight a central point and say there is a temporal depth collapse I think it allows us to do something with it in clinical work and in diagnostic work and that's my hope it may not materialize but this is what I'm hoping for so well even from a researcher's ethological perspective it's kind of like that's the transferable and the interesting research question how could you be observing not on the inside and make some kind of inference or assessment that would matter clinically with no map territory pretense not that it's 100% reliability just taking a clear look at Diagnostics available rethinking it from dynamical systems perspective maybe different uh Biometrics maybe pupil it's like it's not it's just kind of that's the open question and then where it provides clinical utility that's where it matters yeah thank you perfectly said yeah I agree okay thank you Alexi see you thank you Daniel see you byebye e